



Management of Structures

**Maintenance and
Repair**

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

The useable life of a mining machine is often governed by the integrity of the frame and/or other major structures. By introducing a program to effectively manage these structures, there is a potential to significantly extend the life of the equipment, which in turn allows sites to defer capital expenditure and ultimately reduce the Total Cost of Ownership (TCO) of their equipment.

This Best Practice provides a high-level overview of a complete management of structures program that focuses on the process to support the identification, tracking and repair of structural defects within structures. This acute focus on locations and sources of excessive stresses can support all parties within the supply chain.

2.0 Best Practice Description

The development of an effective Management of Structures program requires a clear understanding of the contributors and responsible parties that will ultimately influence the serviceable life of a major structure:

1. Machine Design

The manufacturer is responsible to design a structure will include economical, and manufacturability criteria. The overarching expectation for structure design is to meet or exceed the design life based off the loads and stresses that the machine will be subjected to under typical operations within a mining operational environment.

2. Machine Quality

The manufacturer is required to assemble and weld the structures to meet quality standards that are defined by industry and/or within the business where appropriate.

3. Operational Envelope

The manufacturer will provide operational manuals that specify the operating conditions for which the machine has been designed. This is often supported though onboard sensors and alarms to identify where a threshold has been exceeded along with ongoing operator training and coaching.

The customer should endeavor to operate the machine as per the provided guidance; however, they may still experience events that exceed the designed stress limitations of the structures. Events such as inadvertent contact with the stops when curling a bucket, contacting the face or other objects while still slewing, haul road condition, overloading, uncontrolled face movement etc. All these examples may have a negative influence the structure durability.

Operate for reliability (O4R) programs are highly recommended as part of a structural management program. Machine reliability is significantly impacted by how the machine is operated, therefore, identification and monitoring of the alarms that are generated by incorrect

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operating conditions/technique provides a valuable input for coaching purposes and the ability to minimize stresses within machines.

4. Condition Monitoring

A good condition monitoring and inspection program is recommended to ensure that any cracks that are developing within frames are identified as early as possible. This allows for repairs to be managed before the magnitude of the task becomes significant.

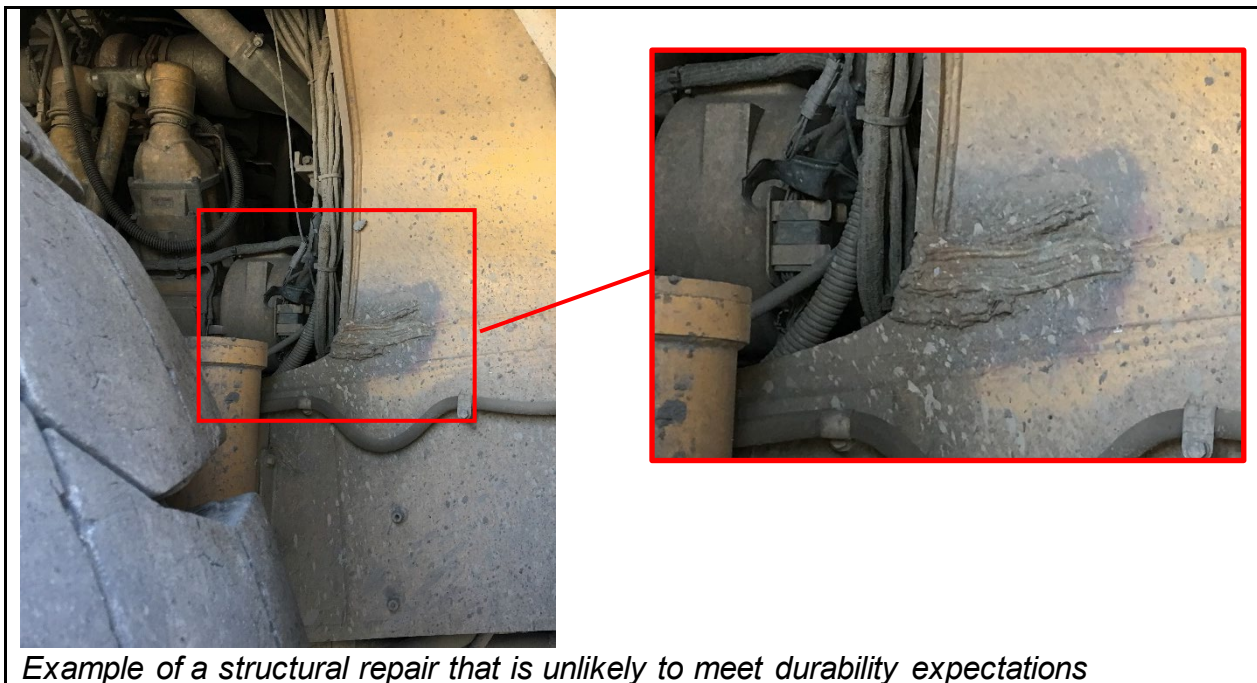
The Operation and Maintenance Manual (O&MM) for Caterpillar mining equipment provides recommendation for the frequency for which the frame should be inspected. This inspection guidance also identifies specific areas should be inspected on a regular basis due to known stresses and loading in the equipment.

5. Repair Management

Once a defect (crack or potentially mechanically induced damage) has been identified within the structure it is critical to ensure that a quality repair is performed as soon is reasonably practicable.

Whoever conducts this repair should:

- Be appropriately trained with demonstrated competency
- Have the correct tools and setup for the task
- Follow repair instructions for specific repairs as provided by the manufacturer
- Have an adequate Quality Assurance (QA) process / system in place to verify repair quality



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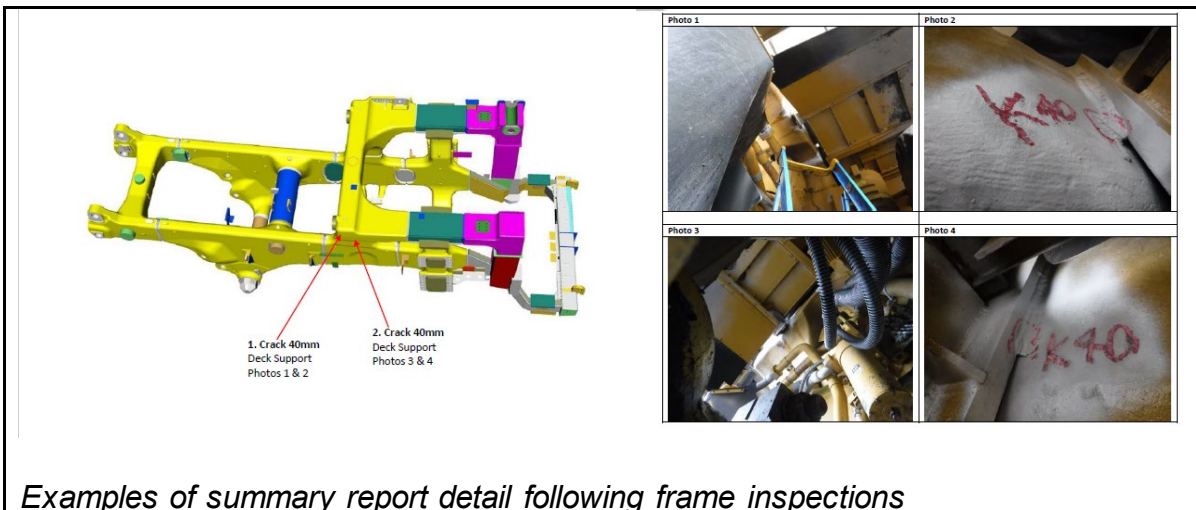
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6. Current Process Gaps

The current inspection sheets that are used by many customers will provide a snapshot of the frame condition and feed into their repair processes. When challenged however, many customers will struggle to identify details that could benefit their current programs.

When evaluating the current situation for your site are you able to answer the following questions:

- Where exactly are the cracks identified on your equipment?
- Is cracking limited to a few select machines or all similar equipment in the fleet?
- How frequently are repairs being performed?
- What is the status and rate of change of the defect? E.g. new, growing, doubled in a week.
- Which areas were inspected during last PM?
- Do inspectors know what areas to inspect?
- What criteria defines when a repair is required?
- What is the required repair procedure?
- Are there factory updates available to address problematic areas?
- Is it more cost effective to repair reactively? Invest in updates? Or should site consider replacing that piece of equipment?
- Were the previous repairs as per OEM recommendations?



3.0 Implementation Steps

1. Detailed Inspection Document with Locations

The frames should be examined in detail and where possible each area of the machine be assigned a unique zone identification and where possible each weld location on the frame be individually identified. This naming structure allows for further customization and addition of inspection locations as required throughout the life of the equipment.

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A numbering system that has been used historically results in a 10-digit location identification as per the following:

Location Identification Structure: CCCC-ZZPP-IP

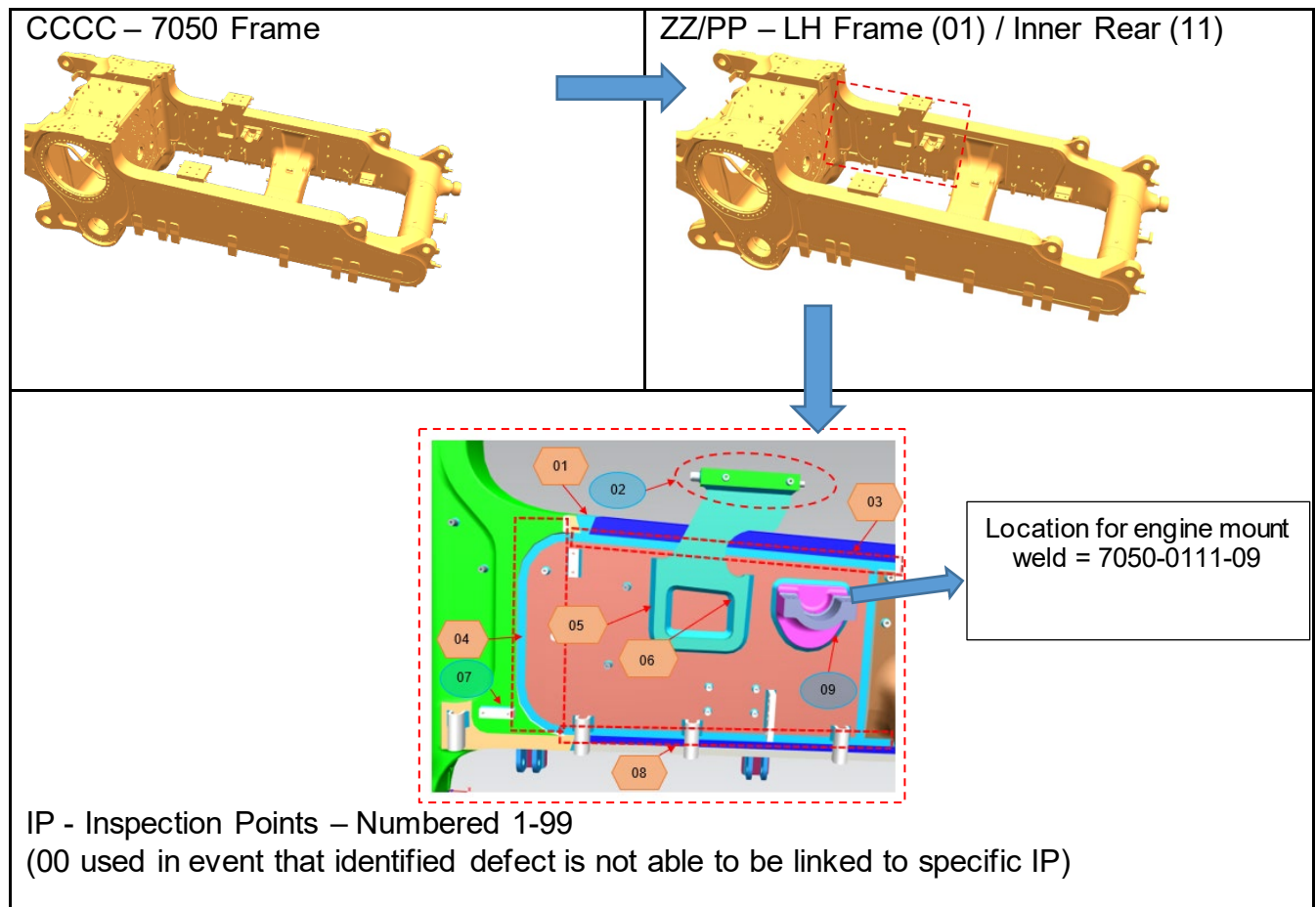
CCCC – 4 Digit Service Management Control System (SMCS) Code – Note that for majority of frames this will be: 7050

ZZ – Zone - High level breakdown of frames into zones, such that every part of the structure fits into a zone.

PP – Position - Should further breakdown of the zones be required this can be used to identify more specifically where in the zone that inspection is reviewing

IP – Inspection point – This is a specific area within that frame that is being reviewed This can be a weld, casting area, mounting block or any unique feature that may exhibit cracking throughout the life of the equipment.

There are inherent complexities in a numbering system that will cover all types of equipment however the intent is to create a database for each model that then allows tracking of all zones / locations across the machine.



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This numbering system can be used to build a highly detailed inspection document that will be foundational to the program.

Examples of documents can be found at within the Best Practice Sharepoint. (Link below)

<https://caterpillar.sharepoint.com/teams/MiningBestPractices/Supporting%20Documentation/Forms/AllItems.aspx?RootFolder=%2Fteams%2FMiningBestPractices%2FSupporting%20Documentation%2FManagement%20of%20Structures&FolderCTID=0x012000040103C2B9B57644A4E9807901B407B9>

There is growing demand for these documents and will be added to this storage location as they are developed.

The documents should also include:

- Repair guidance – This guidance is intended to provide the maintenance teams with information to understand whether they need to respond immediately or defer a repair to a more convenient opportunity (such as when components are removed for access).

Normal A [mm]	Monitor B [mm]	Abnormal C [mm]	Critical D [mm]
0	10	20	30

Sites will need to develop their own limits to populate these fields (e.g. using historical data, identified priority and site experience). The initial site limits can be refined as site experience with specific structures is generated.

NB. Some locations may have 'go / no go' tolerances, e.g. Steering linkages, structural bolts, etc.

- Location – Name for the specific inspection point (IP)
- Unique Location Identifier – Database coding for each Inspection Point using the CCCC/ZZPP/IP coding as referenced previously – If a specific location is not identified 00 general should be used.
- Crack Status – A common coding system to understand the nature of the cracking is recommended per the following:
(N) New, (ES) Existing-static, (EG) Existing-growing, (NR) New Repaired, (ER) Existing Repaired]
- Length (mm) – Total length of cracking observed during inspection
- Fault Code – As each of the IP's are inspected they should be documented using the following guidance.
(NCD) – No Crack Detected, (K) – Crack, (INT) – Intermittent Crack, (MD) – Mechanical Damage

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- Priority areas – Using the guidance from the recommended frame inspection documentation these Inspection Points should be uniquely identified and prioritised – This is achieved both by priority shape and reference to the priority in the table below (1= high / 4 = low)
- Weld Procedure - For the majority of weld crack repairs on Caterpillar products the repairs can be performed using the guidance within REHS1841. If however the crack is propagating into parent material or it is a more complex repair there may be specific repair guidance already published by the manufacturer (Refer to SIS for Caterpillar procedures).

In the event of a unique welding procedure needing to be developed. It is recommended that this be developed in consultation with the OEM or if unavailable third party subject matter experts could be considered.

Weld repairs should not be performed without an appropriate procedure that details the materials, techniques, and competencies required to perform the task.

- Picture Reference – Ideally this will align with each site's storage process for photos supporting repairs.
- Initials – Quality control reference where inspector verified the condition for a given inspection point.

A detailed extract for an inspection Zone/Position with Inspection Points table is shown below.

LH FRAME Inner/Rear Section – (7050-0111)

Locations to be inspected for cracks:

Normal A [mm]	Monitor B [mm]	Abnormal C [mm]	Critical D [mm]	Location	Unique Location Identifier	Crack Status	Length [mm]	Fault Code	CAT. Priority	CAT. Weld Procedure	Pic #	Initial
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0	10	20	30	LH Frame – Inner/Rear Section General	7050-0111-00								
0				LH Top Rail Weld Rear	7050-0111-01				1	M0074160			
0				Fender Mount Top & Bosses	7050-0111-02				4	REHS1841			
0				Inner LH Frame Rear/Top Box Section	7050-0111-03				1	REHS1841			
0				Inner LH Frame Rear /Case Section	7050-0111-04				1	REHS1841			
0				LH Frame Fender Mount Outer	7050-0111-05				1	REHS1841			
0				LH Frame Fender Mount Inner	7050-0111-06				1	REHS1841			
0				Mounting Block	7050-0111-07				4	REHS0541			
0				Inner LH Frame Rear/Bottom Box Section	7050-0111-08				1	REHS1841			
0				LH Engine Mount	7050-0111-09				4	REHS1841			
0				LH Mounting Block	7050-0111-10				4	REHS1841			
0				LH Inner Rear Bosses	7050-0111-11				4	REHS1841			
0				LH Mounting Block	7050-0111-12				4	REHS1841			
0				LH Belly Guard Mount #6	7050-0111-13				4	REHS1841			
0				LH Belly Guard Mount #5	7050-0111-14				4	REHS1841			
0				LH Belly Guard Mount #4	7050-0111-15				4	REHS1841			
0				LH Belly Guard Mount #3	7050-0111-16				4	REHS1841			
0				LH Bottom Rail Weld Rear	7050-0111-17				1	M0074160			
Any additional cracks identified (with suggested limits):													

Crack Status: (N) New, (ES) Existing-static, (EG) Existing-growing, (NR) New Repaired, (ER) Existing Repaired

Fault Code: NCD – No Crack Detected, K – Crack, INT – Intermittent Crack, MD – Mechanical Damage

2. Development of inspection schedule for each machine

- Basic visual of high priority areas should be inspected every 500 hrs (or per guidance provided within the machine O&MM).
- Detailed frame inspections should be performed every 6 months utilizing appropriate non destructive testing procedures such as magnetic particle, ultrasonic, dye penetrant or visual – There will be areas on a frame that will remain inaccessible during this inspection.
- Major PCR intervals – With components removed this provides the greatest opportunity to fully evaluate the health of the structures and repair during these events.

3. Database Tracking

Sites will need to develop a database for which the cracks can be documented. As inspections are performed the detail should be captured in the database for further analysis.

This database should maintain a reference between defects identified and the Location Identification Structure

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4. Analysis of Data

As required the information captured within the structured database should be reviewed. The purpose of these reviews can vary significantly from qualification that existing repairs are effective, identification of problematic areas, upgrade evaluation etc.

At a high level however having robust data will allow sites to verify if frames and major structures are fatiguing and where possible develop strategies to support extended machine life.

Where cracking to the same area across multiple units is detected, analysis should be undertaken to determine root cause of the cracking (e.g. batch defect, weld defect, haul road design, operating procedures, change in environmental conditions, etc.).

4.0 Benefits

The financial benefits of a structural management program will vary dramatically between sites and customers based on high level fleet management strategies. There are four main areas for which benefits of this program will be observed by customers.

1. Safety

Unexpected failure of structures in large mobile plant can result in serious injuries to the operators or others in the vicinity of the equipment. Ensuring that the equipment is fit for purpose should always be the number 1 priority.

2. Capital Avoidance

Extending machine life can provide sites with significant capital avoidance benefits.

Deferring a \$6 million machine purchase with extremely low interest rates (3%) will avoid \$180,000 in interest rates per annum. Taxes and capital incentives may influence the total potential value of this deferral.

3. Improved Availability

Crack repairs are often undertaken in a reactive manner, where the crack, once identified, is hastily “patched” up and the machine is placed back into service without understanding and addressing the root cause behind the crack. This method is perceived to be ‘easier’ and can include repairs from the exposed /outside of the frame without complete removal / repair that may be required from the inside of the structure. In these events sites may see numerous repeated repairs where a single extended shutdown will result in a better repair that will produce increased availability.

Once identified, the crack should be assessed and controlled as required. This may require immediate repair, or planned into the next repair window (e.g. service) with

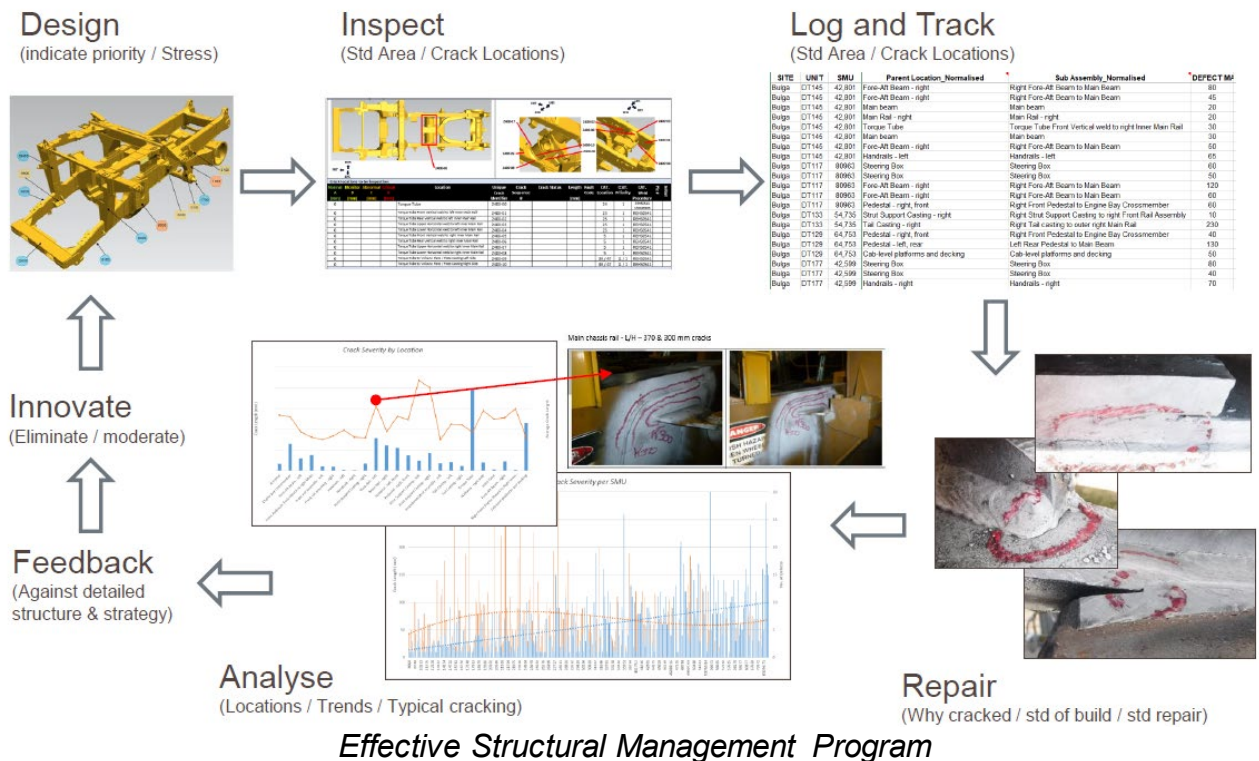
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the aid of a continuous monitoring plan to monitor and assess crack propagation. Where cracking is identified in a significant area(s), the rest of the fleet should be inspected to ensure it is a localized issue to the unit in question.

Should the locations be associated with a potential operational issue the costs of repairs and unavailability may be significant enough to support further technology support (operator assist) or operator training.

4. Effective Communication to Manufacturer

Sharing the structural defect information through to the manufacturers will allow the engineering teams to review the locations for which in field cracking is observed and where required develop updates for the given structures.



5.0 Resources Required

3D Images – The inspection documentation development is best supported by 3D images for each of the given areas however can also be achieved through use of photographs or 2D images.

Tracking Database – No specific recommendations on the database have been provided.

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6.0 Supporting Attachments / References

- Management of Structures – SharePoint Storage

<https://caterpillar.sharepoint.com/teams/MiningBestPractices/Supporting%20Documentation/Forms/AllItems.aspx?RootFolder=%2Fteams%2FMiningBestPractices%2FSupporting%20Documentation%2FManagement%20of%20Structures&FolderCTID=0x012000040103C2B9B57644A4E9807901B407B9>

- Location Identifiers – Attached

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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